

Introduction

- Microplastics are plastic particles smaller than 5 mm that fragment from larger items and accumulate in aquatic ecosystems (Andrady, 2011)
- In freshwater ecosystems, microalgae such as *Scenedesmus quadricauda* serve as critical primary producers; when their growth is inhibited, entire aquatic food webs are disrupted
- Polyethylene terephthalate (PET) is one of the most common plastic polymers in consumer waste and is frequently detected in freshwater systems; PET microplastics can physically interact with algal cells, scatter light, and carry surface-bound additives that may interfere with metabolism
- Freshwater pH fluctuates due to runoff, acid rain, and algal blooms; because photosynthesis and enzyme activity are pH-sensitive, even small pH shifts can alter algal growth rates
- Many studies examine microplastics or pH independently, but far fewer test their combined effects on a model microalga under controlled conditions
- This study investigates how PET microplastics at 100 mg/L interact with pH levels of 7.7 and 8.4 to influence *S. quadricauda* growth, with results informing risk assessment for plastics in freshwater systems experiencing acidification

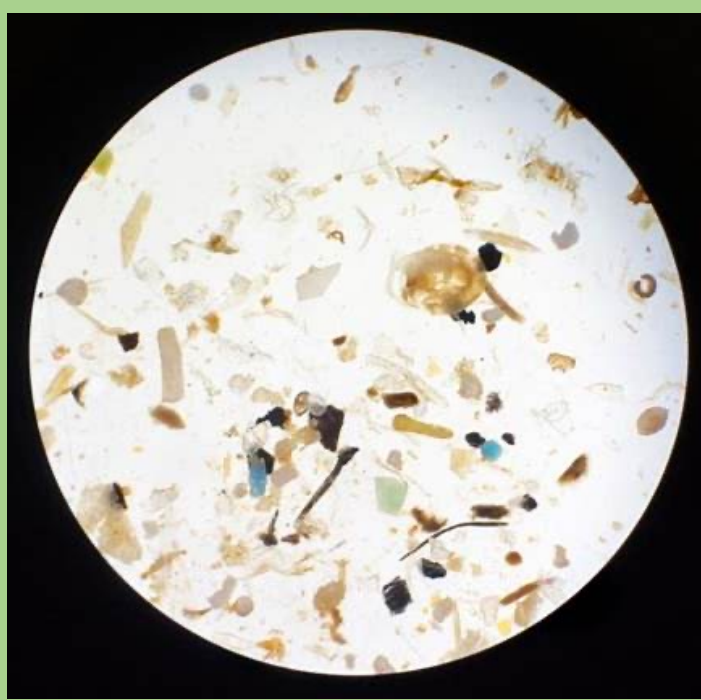


Image: Irvine Ranch Water District, *Microplastics and IRWD*.

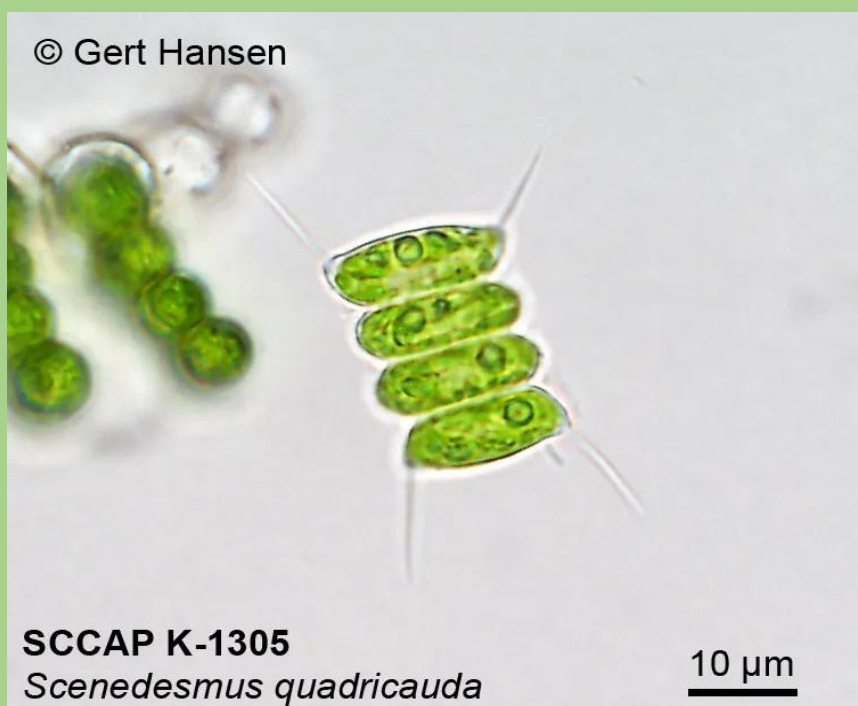


Image: Gert Hansen, *Scenedesmus quadricauda*, Nordic Microalgae, CC license.

Purpose, Hypothesis, & Variables

Purpose: To determine whether PET microplastics at 100 mg/L inhibit the growth of *Scenedesmus quadricauda*, and whether that effect is amplified at a lower pH (7.7 vs. 8.4).

Hypothesis: If *S. quadricauda* is exposed to PET microplastics at 100 mg/L, algal biomass will be lower than in algae-only controls, and this reduction will be greater at pH 7.7 than at pH 8.4, indicating a synergistic stressor interaction.

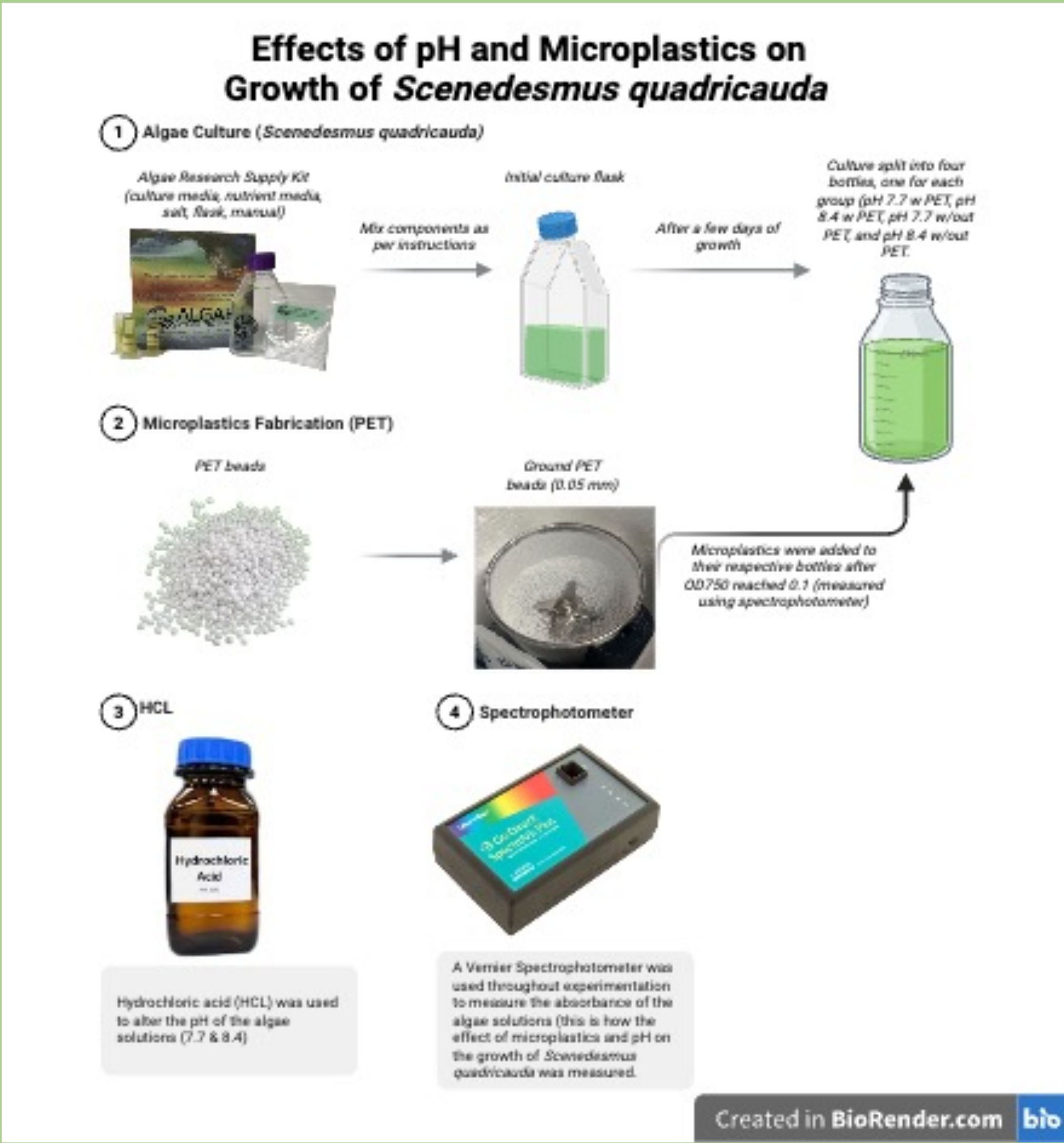
Independent Variable: Treatment group — pH level (7.7 or 8.4) crossed with presence or absence of PET microplastics at 100 mg/L

Dependent Variable: Algal biomass, measured as corrected OD750 (Raw OD750 – Blank OD750) over 14 days

Controls: pH 8.4 algae-only control; pH 7.7 algae-only control; plastic-only blanks at each pH (no algae) to correct for microplastic turbidity

Effects of pH and Microplastics on Growth of Scenedesmus quadricauda

Materials & Methods



Culture & Setup

- *Scenedesmus quadricauda* culture kit (Algae Research Supply) and spring water
- Six 500 mL flasks inoculated to OD750 ≈ 0.10; 150 mL per flask

Microplastics Fabrication

- 3D-printed PET parts ground with a dedicated spice grinder and sieved to 50-100 μm

Experimental Groups

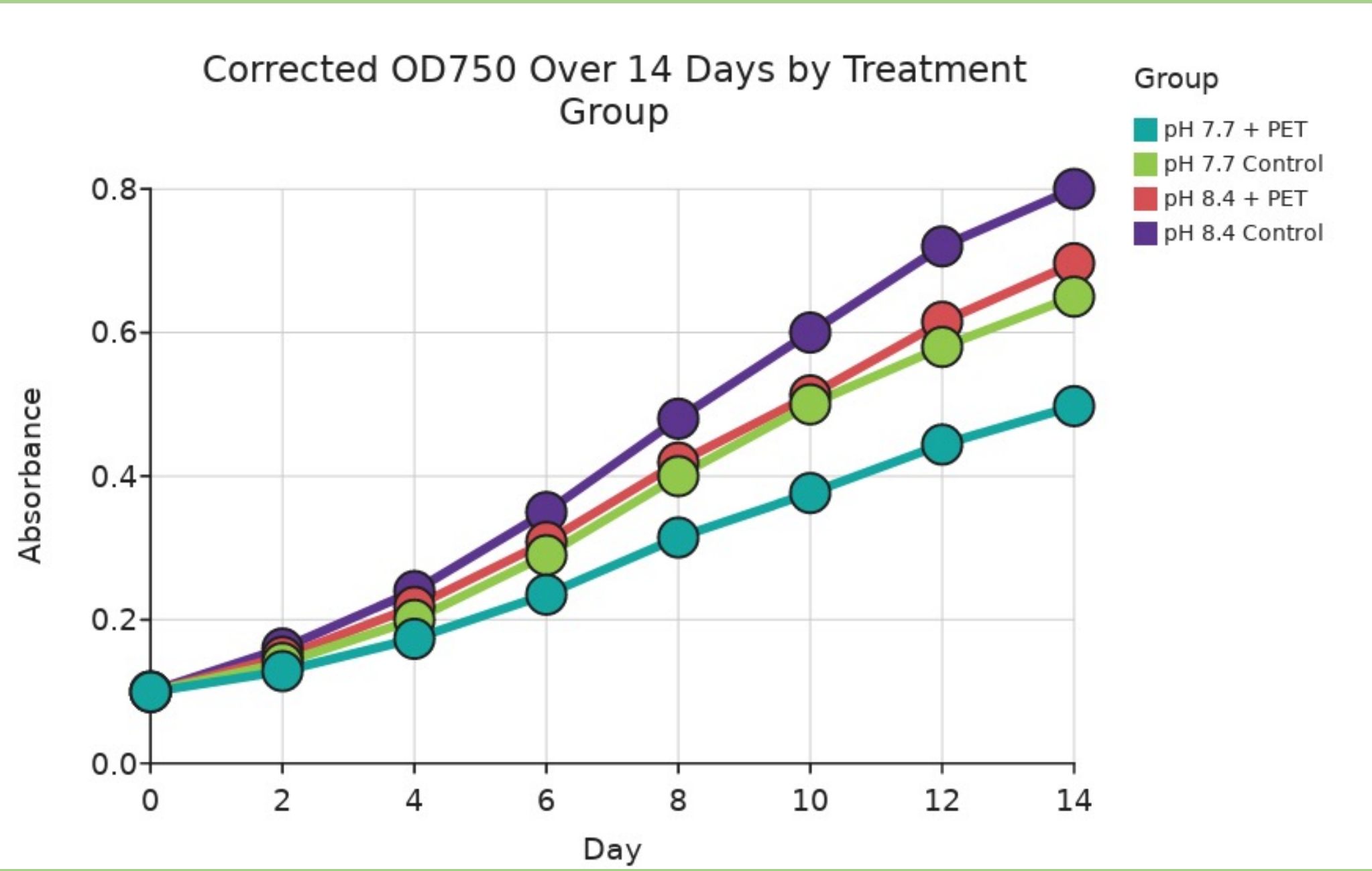
- pH 8.4 Control w/out PET (1 flask)
- pH 8.4 + PET at 100 mg/L (2 flasks)
- pH 7.7 Control w/out PET (1 flask)
- pH 7.7 + PET at 100 mg/L (2 flasks)

Measurement

- OD750 measured with Vernier GoDirect SpectroVis Plus every 2 days (Days 0–14)
- Corrected OD750 = Raw OD750 – Blank OD750 (plastic-only blank, matched by pH)
- pH checked each measurement day; adjusted with 0.1 M HCl or NaOH if needed

**Entire method done twice for two rounds of data collection*

Results



Round 1 — CORRECTED OD750 (raw – blank)							
Date	Day	pH 8.4 Control	pH 8.4 +PET R1	pH 8.4 +PET R2	pH 7.7 Control	pH 7.7 +PET R1	pH 7.7 +PET R2
Dec 20	0	0.100	0.100	0.099	0.100	0.100	0.099
Dec 22	2	0.160	0.150	0.147	0.140	0.130	0.127
Dec 24	4	0.240	0.221	0.215	0.200	0.176	0.171
Dec 26	6	0.350	0.312	0.305	0.290	0.238	0.232
Dec 28	8	0.480	0.423	0.415	0.400	0.318	0.311
Dec 30	10	0.600	0.520	0.505	0.500	0.381	0.372
Jan 1	12	0.720	0.622	0.608	0.580	0.448	0.440
Jan 3	14	0.800	0.705	0.688	0.650	0.503	0.492

Round 2 — CORRECTED OD750 (raw – blank)							
Date	Day	pH 8.4 Control	pH 8.4 +PET R1	pH 8.4 +PET R2	pH 7.7 Control	pH 7.7 +PET R1	pH 7.7 +PET R2
3/4	0	0.100	0.102	0.101	0.102	0.107	0.103
3/6	2	0.166	0.154	0.149	0.151	0.131	0.126
3/10	4	0.243	0.227	0.219	0.213	0.171	0.161
3/12	6	0.337	0.303	0.296	0.297	0.224	0.219
3/16	8	0.452	0.404	0.395	0.382	0.298	0.293
3/18	10	0.570	0.499	0.488	0.452	0.346	0.332
3/24	12	0.628	0.560	0.548	0.506	0.388	0.327
3/26	14	0.692	0.618	0.606	0.557	0.427	0.416

ANCOVA results showing significant effects of treatment group, day, and their interaction on corrected OD750

Effect	Degrees of Freedom (df)	F-statistic (MS / MS residual)	P-value	Interpretation of P
Group (Z)	3	60	<0.01	Means: extremely strong confidence that there are differences between groups
Day (X)	1	2400	<0.01	Means: extremely strong confidence that there is a significant relationship
Interaction (X * Z)	3	31	<0.01	Means: extremely strong confidence that the effect of X depends on Z
Error or Residual	24			

Conclusions

- All four treatment groups exhibited growth over 14 days, but final biomass differed substantially by condition: the pH 8.4 control achieved the highest biomass (OD750 ≈ 0.80), while the pH 7.7 + PET group showed the lowest (OD750 ≈ 0.50)
- PET microplastics at 100 mg/L significantly inhibited *S. quadricauda* growth at both pH levels: the pH 8.4 + PET group showed ~12% lower biomass than its control, supporting the hypothesis
- Lower pH (7.7) amplified the negative effect of PET, with the pH 7.7 + PET group showing ~23% lower growth than its own control and ~37% lower than the pH 8.4 control
- ANCOVA confirmed statistically significant differences between all groups and a significant group × day interaction (F = 31, p < 0.01), indicating that pH and PET interact to produce different growth rates
- Reduced primary producer biomass can cascade into decreased oxygen production, disrupted nutrient cycling, and broader food web instability
- Future work should investigate additional polymer types, a wider pH range, longer exposure durations, and additional measurements such as chlorophyll content and photosynthetic efficiency



Image: Ivan Radic, *Plastic bottles polluting the river*, Wikimedia Commons, CC BY 2.0.

References

- Cole, M. (2011). *Microplastics as a pollutant in the marine environment: A review*. Marine Pollution Bulletin, 62(12), 2588-2597. <https://doi.org/10.1016/j.marpolbul.2011.09.025>
- Wu, Y. (2019). *Effect of microplastics exposure on the photosynthesis system of freshwater algae*. J Hazard Mater. 10.1016/j.jhazmat.2019.04.039
- Reynolds, C. S. (2006). *The ecology of phytoplankton*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511542145>
- Guinotte, J. (2008). *Ocean Acidification and Its Potential Effects on Marine Ecosystems*. Annals of the New York Academy of Sciences, 1134: 320-342. <https://doi.org/10.1196/annals.1439.013>
- Andrady, A. L. (2011). Microplastics in the marine environment. *Marine Pollution Bulletin*, 62(8), 1596–1605.